

Ossp PIPES

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```
1. #include <stdio.h>
#include <sys/types.h>
#include <sys/stat.h>

int main()
{
    if (mkfifo("myfifo", 0666) == -1)
    {
        perror("mkfifo");
        return 1;
    }

    printf("FIFO created\n");

    return 0;
}
```

2. FIFO_READ

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[100];

    fd = open("myfifo", O_RDONLY);

    read(fd, buffer, sizeof(buffer));

    printf("Received: %s\n", buffer);

    close(fd);

    return 0;
}
```

```
}
```

3. FIFO_write.c

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main()
{
    int fd;
    char message[] = "Hello from Writer";

    fd = open("myfifo", O_WRONLY);

    write(fd, message, strlen(message) + 1);

    close(fd);

    printf("Message sent\n");

    return 0;
}
```

Solutions

```
monish_putti@monhh: ~/ossp_homework/message_passing
monish_putti@monhh:~$ mkdir ossp_homework
monish_putti@monhh:~$ cd ossp_homework
monish_putti@monhh:~/ossp_homework$ mkdir message_passing
monish_putti@monhh:~/ossp_homework$ cd message_passing
monish_putti@monhh:~/ossp_homework/message_passing$ pwd
/home/monish_putti/ossp_homework/message_passing
monish_putti@monhh:~/ossp_homework/message_passing$ nano msg_sender.c
monish_putti@monhh:~/ossp_homework/message_passing$ nano msg_receiver.c
monish_putti@monhh:~/ossp_homework/message_passing$ nano msg_fifo.c
monish_putti@monhh:~/ossp_homework/message_passing$ gcc msg_fifo.c -o msg_fifo
monish_putti@monhh:~/ossp_homework/message_passing$ gcc msg_sender.c -o msg_sender
monish_putti@monhh:~/ossp_homework/message_passing$ gcc msg_receiver.c -o msg_receiver
monish_putti@monhh:~/ossp_homework/message_passing$ ./msg_fifo
[INFO] Named pipe created successfully: mypipe
monish_putti@monhh:~/ossp_homework/message_passing$ ./msg_receiver (Terminal 1 - Receiver)
[RECEIVER] Waiting for messages on FIFO (mypipe)...
monish_putti@monhh:~/ossp_homework/message_passing$ ./msg_sender (Terminal 2 - Sender)
Enter message to send: Hello from Sender!
[SENDER] Message sent: Hello from Sender!
Enter message to send: This is OSSP Project - FIFO
[SENDER] Message sent: This is OSSP Project - FIFO
Enter message to send: Process Communication using Named Pipe
[SENDER] Message sent: Process Communication using Named Pipe
Enter message to send: exit
[SENDER] Exiting sender.
[RECEIVER] Received: Hello from Sender!
[RECEIVER] Received: This is OSSP Project - FIFO
[RECEIVER] Received: Process Communication using Named Pipe
[RECEIVER] Receiver terminated.
monish_putti@monhh:~/ossp_homework/message_passing$ ls -l
total 72
-rwxr-xr-x 1 monish_putti monish_putti 16048 Sep 28 20:15 msg_fifo
-rwxr-xr-x 1 monish_putti monish_putti 16136 Sep 28 20:16 msg_receiver
-rwxr-xr-x 1 monish_putti monish_putti 16168 Sep 28 20:16 msg_sender
-rw-r--r-- 1 monish_putti monish_putti 964 Sep 28 20:15 msg_fifo.c
-rw-r--r-- 1 monish_putti monish_putti 1223 Sep 28 20:16 msg_receiver.c
-rw-r--r-- 1 monish_putti monish_putti 1205 Sep 28 20:16 msg_sender.c
monish_putti@monhh:~/ossp_homework/message_passing$
```